9.2 Populations, Samples, and Hypotheses

Name _____ Period ____ Date _____

Algebra II 9.2 (30-minute plan) · Student sheet

I can…

- tell the difference between a population and a sample, and between a parameter and a statistic.
- use a simulation to find proportions.
- use a simulation to decide whether to reject a hypothesis.

Vocabulary · fill in the missing word

Population — the entire group you want _____ about

Sample — a part of the _____

Parameter — a number that describes a _____

Statistic — a number that _____ a sample

Hypothesis — a claim about a _____ of a population

Simulation — a model of a _____ situation, often using random numbers or coins

Part 1 · Notes and examples (fill in each note, then work the example)

Population: _____. **Sample:** _____.

Example 1

A school surveys 100 of its 1,200 students about the lunch menu. Identify the population and the sample.

Parameter or statistic: A **parameter** describes a _____; a **statistic** describes a _____.

Example 2

Tell whether each number is a parameter or a statistic.

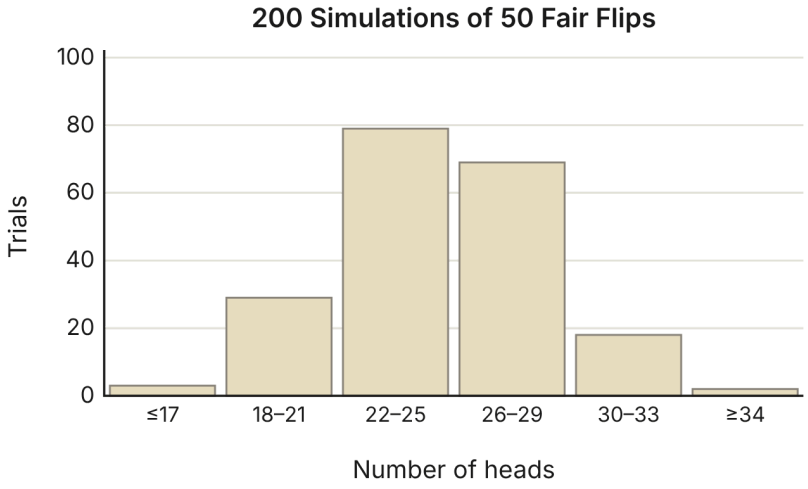
a. The mean age of all 30 members of a club is 34.

b. In a survey of 1,000 voters, 52% plan to vote yes.

Testing a hypothesis: Reject a hypothesis when the result is very _____ if the hypothesis is true.

Example 3

You claim a coin is fair. In 50 flips, you get 34 heads. A simulation of 200 trials of 50 fair coin flips gave 34 or more heads in only 2 trials. What should you conclude?



Part 2 · Partner practice

Partner practice

Tell whether each data set is a population or a sample.

a. the ages of every person counted in a town’s census

b. the heights of 30 randomly chosen 10th graders